Dynamical Mechanisms of Hyperuniformity: A General Framework for Coupled Stochastic Fields

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1 General n-field theory of long-wavelength density fluctuations

The purpose of this framework is to identify the dynamical mechanisms controlling density fluctuations around a homogeneous state. The deterministic spectrum determines stability, but the stationary structure factor also depends on stochastic forcing and on how the excited modes project onto density. All exact statements below refer to the specified linear Gaussian fluctuation theory; they are not claims of exactness for the underlying nonlinear many-body dynamics.

1.1 Stochastic continuum model and controlled linearization

Consider n real fields on a periodic volume $V = L^d$,

$$\mathbf{u} = (\rho, \phi_1, \ldots, \phi_{n-1})^\top, \qquad \partial_t \mathbf{u} = \mathbf{F}[\mathbf{u}] + \sqrt{\varepsilon}\, \mathbf{G}[\mathbf{u}]\boldsymbol{\xi}. \tag{1}$$

Here ρ is a conserved number density, the auxiliary fields may themselves be conserved, and $\mathbf{G}$ can contain spatial derivatives or convolution kernels, $G[\mathbf{u}] \in \mathbb{C}^{n\times p}$. The equation is interpreted in the Itô sense, with p independent real noise channels

$$\boldsymbol{\xi} = (\xi_1, \ldots, \xi_p)^\top, \qquad \langle \xi_a(\mathbf{r}, t)\rangle = 0, \tag{2}$$

satisfying

$$\langle \xi_a(\mathbf{r}, t)\xi_b(\mathbf{r}', t')\rangle = \delta_{ab}\delta(\mathbf{r} - \mathbf{r}')\delta(t - t'). \tag{3}$$

A spatial cutoff is understood whenever required to define the continuum noise. Assume a homogeneous deterministic solution $\mathbf{u}_0$ with $\mathbf{F}[\mathbf{u}_0] = 0$ and $\rho_0 > 0$, and translation-invariant linearized coefficients. Writing $\mathbf{u} = \mathbf{u}_0 + \sqrt{\varepsilon}\,\mathbf{v}$ gives, at leading order,

$$\partial_t \mathbf{v} = \mathcal{L}\mathbf{v} + \mathbf{G}[\mathbf{u}_0]\boldsymbol{\xi}, \qquad \mathcal{L} = \left.\frac{\delta \mathbf{F}}{\delta \mathbf{u}}\right|_{\mathbf{u}_0}. \tag{4}$$

The expansion and its order counting are derived in Appendix 4.1. Thus freezing multiplicative noise at $\mathbf{u}_0$ is a linear-noise approximation justified by the fluctuation expansion, not an exact linearization at arbitrary noise strength. Below $\mathbf{x}$ denotes the physical fluctuation within this approximation; its noise matrix includes $\sqrt{\varepsilon}$ if that parameter is retained. Nonlinear corrections and fluctuation-induced changes of the background are outside the present closure.

Use the volume-normalized Fourier transform

$$\mathbf{x}_\mathbf{k} = V^{-1/2} \int_V e^{-\mathrm{i}\mathbf{k}\cdot\mathbf{r}} \delta\mathbf{u}(\mathbf{r})\, \mathrm{d}\mathbf{r}, \qquad \mathbf{k} = 2\pi\mathbf{m}/L, \quad \mathbf{m} \in \mathbb{Z}^d. \tag{5}$$

Each nonzero wavevector obeys an n-component Ornstein–Uhlenbeck equation,

$$\boxed{\dot{\mathbf{x}}_\mathbf{k} = A(\mathbf{k})\mathbf{x}_\mathbf{k} + \boldsymbol{\eta}_\mathbf{k}, \qquad Q(\mathbf{k}) = B(\mathbf{k})B(\mathbf{k})^\dagger,} \tag{6}$$

where A is the Fourier symbol of $\mathcal{L}$, B is an $n \times p$ matrix, and $\boldsymbol{\eta} = B\boldsymbol{\xi}$. The dagger denotes conjugate transpose. Reality requires $\mathbf{x}_{-\mathbf{k}} = \mathbf{x}_\mathbf{k}^*$, $A(-\mathbf{k}) = A(\mathbf{k})^*$, and $Q(-\mathbf{k}) = Q(\mathbf{k})^*$. In particular, the two equivalent noise conventions are

$$\langle \boldsymbol{\eta}_\mathbf{k}(t)\boldsymbol{\eta}_{\mathbf{k}'}^\dagger(t')\rangle = Q(\mathbf{k})\delta_{\mathbf{k},\mathbf{k}'}\delta(t - t'), \tag{7}$$

$$\langle \boldsymbol{\eta}_\mathbf{k}(t)\boldsymbol{\eta}_{\mathbf{k}'}^\top(t')\rangle = Q(\mathbf{k})\delta_{\mathbf{k},-\mathbf{k}'}\delta(t - t'). \tag{8}$$

The sign in the wavevector delta must therefore change when transpose is replaced by conjugate transpose. Correlated physical noise channels can always be included through a positive-semidefinite Q and a matrix square root B.

1.2 Conservation and the stationary covariance theorem

Let $\mathbf{e}_\rho = (1, 0, \ldots, 0)^\top$. Exact number conservation, $\partial_t \rho = -\nabla \cdot \mathbf{J}$, implies

$$\mathbf{e}_\rho^\dagger A(\mathbf{0}) = 0, \qquad \mathbf{e}_\rho^\dagger B(\mathbf{0}) = 0. \tag{9}$$

Thus conservation supplies a left zero eigenvector and a zero eigenvalue, but the associated right eigenvector need not be the density coordinate vector. Several conservation laws give several such constraints. For a random current with a bounded long-wavelength covariance, $Q_{\rho\rho}(\mathbf{k}) = O(k^2)$; conservation alone does not imply a nonzero k^2 coefficient or constrain singular, spatially correlated currents to this power.

Stationary covariance theorem. For every fixed $\mathbf{k} \neq 0$ in the range of validity of the continuum theory, assume that $A(\mathbf{k})$ is Hurwitz:

$$\max_j \operatorname{Re} \lambda_j[A(\mathbf{k})] < 0. \tag{10}$$

Then the linear process has a unique attracting stationary Gaussian law (possibly degenerate), whose Hermitian positive-semidefinite covariance is

$$C(\mathbf{k}) = \langle \mathbf{x}_\mathbf{k} \mathbf{x}_\mathbf{k}^\dagger \rangle = \langle \mathbf{x}_\mathbf{k} \mathbf{x}_{-\mathbf{k}}^\top \rangle, \tag{11}$$

$$C(\mathbf{k}) = \int_0^\infty e^{A(\mathbf{k})t} Q(\mathbf{k}) e^{A(\mathbf{k})^\dagger t} \, \mathrm{d}t, \tag{12}$$

$$AC + CA^\dagger + Q = 0. \tag{13}$$

Indeed, the finite-time covariance is

$$C(t) = e^{At} C(0) e^{A^\dagger t} + \int_0^\top e^{As} Q e^{A^\dagger s} \, \mathrm{d}s. \tag{14}$$

Exponential decay at fixed $\mathbf{k}$ removes the initial term and makes the integral converge. Differentiating its integrand proves Eq. (13). If two solutions differed by H, then $e^{At} H e^{A^\dagger t} = H$ for all t, so decay implies $H = 0$. Positive semidefiniteness follows directly from the integral; positive definiteness additionally requires that the noise excite every state direction through the dynamics. No diagonalizability or normality assumption is needed.

The Hurwitz condition ensures an attracting stationary law for arbitrary finite initial covariance. Without it, unforced neutral or unstable directions can admit exceptional invariant measures, but these do not establish a robust stationary fluctuation theory around a stable state. At exactly $\mathbf{k} = 0$, conservation precludes the above Hurwitz assumption. Fixed particle number sets $\delta\rho_\mathbf{0} = 0$; this constraint is not evidence of hyperuniformity.

1.3 Exact density criterion and frequency representation

With the Fourier normalization above, the dimensionless structure factor is

$$\boxed{S(\mathbf{k}) = \rho_0^{-1} \mathbf{e}_\rho^\dagger C(\mathbf{k}) \mathbf{e}_\rho.} \tag{15}$$

For point particles this agrees, at nonzero allowed wavevectors, with $N^{-1}\langle |\sum_j e^{-\mathrm{i}\mathbf{k}\cdot\mathbf{r}_j}|^2 \rangle$. Hyperuniformity is the limit $S(\mathbf{k}) \to 0$ through nonzero wavevectors in the thermodynamic limit at fixed ρ_0. For anisotropic systems this limit must hold along every approach to the origin; radial averaging alone does not establish it.

Equations (12)–(15) give the following necessary and sufficient criterion within the linear theory:

$$\boxed{\text{hyperuniformity} \iff \lim_{\mathbf{k}\to 0} \int_0^\infty \| \mathbf{e}_\rho^\dagger e^{A(\mathbf{k})t} B(\mathbf{k}) \|_2^2 \, \mathrm{d}t = 0.} \tag{16}$$

This nonnegative expression includes all interference and transient amplification. With temporal convention $\mathbf{x}(t) = \int e^{-\mathrm{i}\omega t} \widehat{\mathbf{x}}(\omega) \, \mathrm{d}\omega/(2\pi)$, define

$$R(\mathbf{k}, \omega) = (-\mathrm{i}\omega I - A(\mathbf{k}))^{-1}. \tag{17}$$

The equivalent frequency-domain formula is

$$C = \int_{-\infty}^\infty R Q R^\dagger \frac{\mathrm{d}\omega}{2\pi}, \qquad S = \frac{1}{\rho_0} \int_{-\infty}^\infty \| \mathbf{e}_\rho^\dagger R B \|_2^2 \frac{\mathrm{d}\omega}{2\pi}. \tag{18}$$

Stationarity is taken first for each nonzero mode. Neither interchange of the long-time and long-wavelength limits nor interchange with a singular parameter limit follows from stability at fixed $\mathbf{k}$.

1.4 Mode-resolved formula and conditional scaling laws

When $A(\mathbf{k})$ is diagonalizable at fixed nonzero $\mathbf{k}$, choose biorthogonal right and left eigenvectors,

$$A\mathbf{r}_j = \lambda_j \mathbf{r}_j, \qquad \mathbf{l}_j^\dagger A = \lambda_j \mathbf{l}_j^\dagger, \qquad \mathbf{l}_i^\dagger \mathbf{r}_j = \delta_{ij}. \tag{19}$$

Define $h_j = \mathbf{e}_\rho^\dagger \mathbf{r}_j$ and $q_{ij} = \mathbf{l}_i^\dagger Q \mathbf{l}_j$. Substitution of $e^{At} = \sum_j e^{\lambda_j t} \mathbf{r}_j \mathbf{l}_j^\dagger$ into Eq. (12) yields

$$\boxed{S(\mathbf{k}) = -\frac{1}{\rho_0} \sum_{i,j=1}^{n} \frac{h_i q_{ij} h_j^*}{\lambda_i + \lambda_j^*}.} \tag{20}$$

The denominators encode relaxation and relative propagation frequencies; q_{ij} encodes noise excitation and correlations; $h_i h_j^*$ encodes density content. The slowest eigenvalue alone is insufficient. Individual cross terms need not be positive, although their complete sum is real and nonnegative. At defective points one must use Eq. (12) or (18); Jordan blocks generate polynomial factors in t that can alter scaling.

Single-mode scaling theorem. For an exact scalar closure $\dot{x} = (-\Gamma(k) + \mathrm{i}\Omega(k))x + \eta$ with $\Gamma > 0$ and noise covariance $q(k)\delta(t-t')$, its stationary variance is $q/(2\Gamma)$. If density equals $h(k)x$, and

$$\Gamma \sim g k^z, \qquad q \sim q_* k^\beta, \qquad |h|^2 \sim H k^{2\gamma}, \qquad g, q_*, H > 0, \tag{21}$$

then

$$S(k) \sim \frac{H q_*}{2\rho_0 g} k^{\beta + 2\gamma - z}. \tag{22}$$

The same conclusion holds for a reduced scalar description only if the omitted contribution to the density covariance is of smaller order than this expression. For $h = 1$, this is the familiar $\beta - z$ rule. Dean noise gives $q = 2D\rho_0 k^2$: ordinary diffusion has $\Gamma = Dk^2$ and $S = 1$, whereas a valid scalar model with $\Gamma \to g > 0$ has $S \sim (D/g)k^2$. A finite limiting rate at nonzero k is incompatible with a regular local scalar conserved drift at $k = 0$; it requires, for example, a singular nonlocal kernel. Conservation by itself does not justify such a closure.

Multi-mode scaling theorem with explicit hypotheses. Set $\Gamma_j = -\operatorname{Re}\lambda_j > 0$ and

$$d_j = \frac{|h_j|^2 q_{jj}}{2\rho_0 \Gamma_j}, \qquad S_{\mathrm{diag}} = \sum_j d_j, \qquad S_{\mathrm{cross}} = S - S_{\mathrm{diag}}. \tag{23}$$

Assume finitely many modes and positive leading power laws for all nonzero d_j:

$$\Gamma_j \sim g_j k^{z_j}, \quad q_{jj} \sim q_j k^{\beta_j}, \quad |h_j|^2 \sim H_j k^{2\gamma_j}, \qquad g_j, q_j, H_j > 0. \tag{24}$$

If $q_{ij} = 0$ for $i \neq j$, or more generally $S_{\mathrm{cross}} = o(S_{\mathrm{diag}})$, then

$$\boxed{S(k) \sim K k^\nu, \qquad \nu = \min_{j : d_j \not\equiv 0} (\beta_j + 2\gamma_j - z_j), \qquad K > 0.} \tag{25}$$

The proof is the finite sum of positive diagonal terms; K is the sum of their leading coefficients at the minimal exponent. All contributing modes, including gapped modes, must be included. The separate exponents of q_{jj} and $|h_j|^2$ depend on eigenvector normalization, but their product and the exponent in Eq. (25) do not. Here ν is used to avoid confusion with the chemotactic sensitivity α.

A useful sufficient criterion survives without neglecting cross terms. Define the noise-channel row function $f_j(t) = h_j e^{\lambda_j t} \mathbf{l}_j^\dagger B$. Cauchy–Schwarz in time and noise space gives

$$0 \le S = \rho_0^{-1} \| \sum_j f_j \|_{L^2}^2 \le \left(\sum_j \sqrt{d_j} \right)^2 \le n \sum_j d_j. \tag{26}$$

Consequently $d_j \to 0$ for every mode is sufficient for hyperuniformity even with correlated modal noise. The converse and the minimum-exponent equality need not hold because of cancellations. Singular eigenvectors, degeneracy, and non-normal amplification must be evaluated using the full covariance rather than an unconditional exponent rule.

Under the hypotheses of Eq. (25), $\nu > 0$ implies hyperuniformity, $\nu = 0$ a finite positive plateau, and $\nu < 0$ divergent long-wavelength fluctuations. Logarithmic or non-power-law spectra must instead be classified by Eq. (16). In an infinite system, a divergent spectrum must also be locally integrable for the claimed stationary field to have finite infrared variance (for an isotropic power law, this requires $\nu > -d$).

1.5 Exact field elimination and conditions for a Markov approximation

Partition the fields into retained variables x (including density) and auxiliary variables y, suppressing $\mathbf{k}$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} A_{ss} & A_{sf} \\ A_{fs} & A_{ff} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} \eta_s \\ \eta_f \end{pmatrix}. \tag{27}$$

If A_{ff} is Hurwitz, solving the second equation from the remote past gives the exact stationary equation

$$\dot{x}(t) = A_{ss}x(t) + \int_0^\infty K(\tau)x(t-\tau)\,\mathrm{d}\tau + \zeta(t), \tag{28}$$

$$K(\tau) = A_{sf}e^{A_{ff}\tau}A_{fs}, \tag{29}$$

$$\zeta(t) = \eta_s(t) + A_{sf}\int_0^\infty e^{A_{ff}\tau}\eta_f(t-\tau)\,\mathrm{d}\tau. \tag{30}$$

Elimination is exact here, but produces a memory kernel and generally colored noise. In frequency space put

$$R_f(\omega) = (-\mathrm{i}\omega I - A_{ff})^{-1}, \tag{31}$$

$$A_{\mathrm{eff}}(\omega) = A_{ss} + A_{sf}R_f(\omega)A_{fs}, \tag{32}$$

$$T(\omega) = \begin{pmatrix} I & A_{sf}R_f(\omega) \end{pmatrix}, \qquad Q_{\mathrm{eff}}(\omega) = T(\omega)QT(\omega)^\dagger. \tag{33}$$

Thus $\widehat{x} = [-\mathrm{i}\omega I - A_{\mathrm{eff}}(\omega)]^{-1}T(\omega)\widehat{\eta}$. This expression retains auxiliary noise and its cross-correlations with the retained noise, and reproduces the retained block of the full covariance through the frequency integral.

The instantaneous approximation replaces $R_f(\omega)$ by $-A_{ff}^{-1}$, giving

$$A_{\mathrm{eff}}^{(0)} = A_{ss} - A_{sf}A_{ff}^{-1}A_{fs}, \qquad Q_{\mathrm{eff}}^{(0)} = T_0QT_0^\dagger, \qquad T_0 = \begin{pmatrix} I & -A_{sf}A_{ff}^{-1} \end{pmatrix}. \tag{34}$$

It is generally incorrect to change the drift while leaving the noise unchanged. A common sufficient starting point for a controlled reduction is a uniformly fast auxiliary semigroup,

$$\|e^{A_{ff}(\mathbf{k})t}\| \le Me^{-gt}, \qquad M, g > 0 \text{ independent of small } \mathbf{k}, \tag{35}$$

together with bounded couplings and retained frequencies small compared with g. More directly, the expansion requires $|\omega|\|A_{ff}^{-1}\| \ll 1$ over the frequencies relevant to density fluctuations. A spectral gap alone does not bound non-normal transient growth or establish this separation. Moreover, a correct low-frequency response is insufficient unless the omitted frequency range contributes a smaller order to the equal-time density covariance. For a proposed reduced spectrum $S_{\mathrm{red}} > 0$, an explicit sufficient check is

$$\int_{-\infty}^{\infty} |P_\rho(\mathbf{k},\omega) - P_{\rho,\mathrm{red}}(\mathbf{k},\omega)|\frac{\mathrm{d}\omega}{2\pi} = o\big(\rho_0 S_{\mathrm{red}}(\mathbf{k})\big), \tag{36}$$

where $P_\rho = \mathbf{e}_\rho^\dagger RQR^\dagger\mathbf{e}_\rho$ and $P_{\rho,\mathrm{red}}$ is the corresponding reduced density spectrum. This implies equality of the leading static asymptotics.

Slow-sector completeness

Every mode contributing to the leading density covariance must be retained, or its memory and noise effects must be represented with an error smaller than that covariance. An auxiliary relaxation rate tending to zero warns that a uniform fast-gap argument fails. Such a mode can still be eliminated in special cases with proven relative time-scale separation or exact decoupling; it cannot be discarded solely by setting its time derivative to zero. Conversely, a finite auxiliary gap permits a reduction only after noise transfer and the covariance error have been checked.

2 Example: chemotactic particles and a dynamical chemical field

2.1 Particle-based model

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{r}_i\left(t\right)=\alpha\nabla c\left(\mathbf{r}_i\left(t\right),t\right)+\sqrt{2D}\boldsymbol{\xi}_i\left(t\right)\ , \tag{37a}$$

$$\frac{\partial}{\partial t}c\left(\mathbf{r},t\right)=D_c\nabla^2 c\left(\mathbf{r},t\right)+\mu c\left(\mathbf{r},t\right)+r\sum_{j=1}^{N}\delta\left(\mathbf{r}-\mathbf{r}_j\left(t\right)\right)\ , \tag{37b}$$

for $i=1,2,\cdots,N$. Here, $\mathbf{r}_i$ is the position of the i-th particle, c is the concentration field of the chemoattractant, α is the chemotactic sensitivity. If $\alpha>0$, the particles are attracted to higher concentrations of the chemoattractant (chemoattraction/positive chemotaxis), while if $\alpha<0$, the particles are repelled by higher concentrations of the chemoattractant (chemorepulsion/negative chemotaxis). D is the diffusion coefficient of the particles, $\boldsymbol{\xi}_i$ is an independent Gaussian white noise with zero mean and unit variance for each particle, D_c is the diffusion coefficient of the chemoattractant, μ is the decay ($\mu<0$) or growth ($\mu>0$) rate of the chemoattractant, and r is the production ($r>0$) or consumption rate ($r<0$) of the chemoattractant by the particles.

For simplicity, we assume that particles are point-like and move in a two-dimensional $L\times L$ square domain with periodic boundary conditions. The initial positions of the particles can be randomly distributed, and the initial concentration field can be set to a small random perturbation around a uniform background concentration c_0.

To reduce the parameter space, we can non-dimensionalize the equations by introducing $x_0=L$, $t_0=1/r$ and c_0 as characteristic scales of length, time and concentration. The resulting dimensionless parameters are $\tilde{\alpha}=\frac{\alpha c_0}{L^2 r}$, $\tilde{D}=\frac{D}{L^2 r}$, $\tilde{D}_c=\frac{D_c}{L^2 r}$, and $\tilde{\mu}=\frac{\mu}{r}$. We can omit the tildes and work with the dimensionless equations:

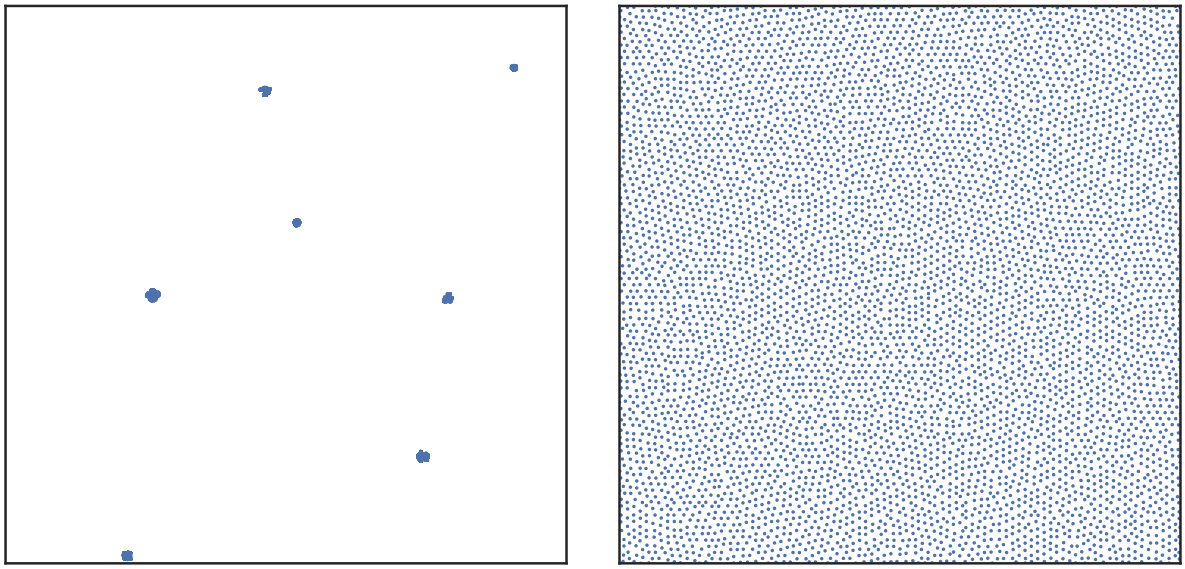
$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{r}_i\left(t\right)=\alpha\nabla c\left(\mathbf{r}_i\left(t\right),t\right)+\sqrt{2D}\boldsymbol{\xi}_i\left(t\right)\ , \tag{38a}$$

$$\frac{\partial}{\partial t}c\left(\mathbf{r},t\right)=D_c\nabla^2 c\left(\mathbf{r},t\right)+\mu c\left(\mathbf{r},t\right)+\sum_{j=1}^{N}\delta\left(\mathbf{r}-\mathbf{r}_j\left(t\right)\right)\ . \tag{38b}$$

Note that the above equations mathematically implies the assumption of $r>0$. If $r<0$, the non-dimensionalization would be different, and the Eq. (38b) would have a negative sign in front of the summation term:

$$\frac{\partial}{\partial t}c\left(\mathbf{r},t\right)=D_c\nabla^2 c\left(\mathbf{r},t\right)+\mu c\left(\mathbf{r},t\right)-\sum_{j=1}^{N}\delta\left(\mathbf{r}-\mathbf{r}_j\left(t\right)\right)\ . \tag{39}$$

In the following, we focus on the case $r<0$, corresponding to chemical consumption. For $\alpha>0$, consumption lowers the chemical concentration around particles and thereby generates an effective repulsion between them, which is more relevant to the suppression of density fluctuations. As shown in the snapshots, the case of production ($r>0$) leads to aggregation of particles and formation of dense clusters, while the case of consumption ($r<0$) leads to dispersion of particles and formation of more uniform distributions. The latter case is more likely to exhibit hyperuniformity.



(a) Production $[\sum_{j=1}^{N} \delta(\mathbf{r} - \mathbf{r}_j(t))]$ (b) Consumption $[-\sum_{j=1}^{N} \delta(\mathbf{r} - \mathbf{r}_j(t))]$

Figure 1: Snapshots of particle configurations for (a) production and (b) consumption of chemoattractant. The parameters are $\alpha = 5$, $D = 10^{-4}$, $D_c = 5$, $\mu = -10^{-4}$, $N = 6400$ and $L = 20$.

2.2 Continuum model

It is convenient to describe collective behavior using a particle density field $\rho(\mathbf{r}, t) = \sum_{i=1}^{N} \delta(\mathbf{r} - \mathbf{r}_i(t))$ (Note that the $\rho(\mathbf{r}, t)$ here is particle number density, not probability density). By applying the Dean's method [1], we can derive the following stochastic partial differential equations for ρ and c:

$$\frac{\partial}{\partial t}\rho(\mathbf{r}, t) = -\alpha\nabla \cdot (\rho\nabla c) + D\nabla^2\rho - \nabla \cdot \left[\sqrt{2D\rho}\mathbf{\Lambda}\right] , \tag{40a}$$

$$\frac{\partial}{\partial t}c(\mathbf{r}, t) = D_c\nabla^2 c + \mu c - \rho , \tag{40b}$$

where $\mathbf{\Lambda}(\mathbf{r}, t)$ is a Gaussian white noise with zero mean and unit variance, satisfying

$$\langle \Lambda_a(\mathbf{r}, t)\Lambda_b(\mathbf{r}', t')\rangle = \delta_{ab}\delta(\mathbf{r} - \mathbf{r}')\delta(t - t') . \tag{41}$$

Under the mean-field approximation, $\langle \mathbf{\Lambda} \rangle = 0$ leads to $\left\langle -\nabla \cdot \left[\sqrt{2D\rho}\mathbf{\Lambda}\right]\right\rangle \approx 0$, and we can neglect the noise term. This gives us the classical Keller-Segel model [2, 3]:

$$\frac{\partial}{\partial t}\rho(\mathbf{r}, t) = -\alpha\nabla \cdot (\rho\nabla c) + D\nabla^2\rho , \tag{42a}$$

$$\frac{\partial}{\partial t}c(\mathbf{r}, t) = D_c\nabla^2 c + \mu c - \rho . \tag{42b}$$

One can perform linear stability analysis on the Keller-Segel model to determine the conditions for pattern formation. However, the mean-field approximation neglects fluctuations, which can be crucial for understanding the emergence of hyperuniformity. Therefore, we need to consider the full stochastic equations with noise to capture the effects of fluctuations on the system's behavior.

One obvious solution of Eqs. (42) is the homogeneous state $\rho(\mathbf{r}, t) = \rho_0$ and $c(\mathbf{r}, t) = c_0$, where $c_0 = \rho_0/\mu$, which can be regarded as solution to Eqs. (40) in the mean-field sense. To analyze the stability of this homogeneous state, we can introduce small perturbations $\delta\rho(\mathbf{r}, t)$ and $\delta c(\mathbf{r}, t)$ around the homogeneous solution:

$$\rho(\mathbf{r}, t) = \rho_0 + \delta\rho(\mathbf{r}, t), \quad c(\mathbf{r}, t) = c_0 + \delta c(\mathbf{r}, t) . \tag{43}$$

Introducing these perturbations into Eqs. (40) gives us the following linearized equations:

$$\frac{\partial}{\partial t}\delta\rho = -\alpha\rho_0\nabla^2\delta c + D\nabla^2\delta\rho - \nabla\cdot\left[\sqrt{2D\rho_0}\mathbf{\Lambda}\right] , \tag{44a}$$

$$\frac{\partial}{\partial t}\delta c = D_c\nabla^2\delta c + \mu\delta c - \delta\rho . \tag{44b}$$

By performing Fourier transform on the linearized equations, we can analyze the growth rates of different modes and determine the conditions for instability. The presence of noise can lead to fluctuations that may drive the system towards hyperuniform states, even in regimes where the mean-field analysis predicts stability. Therefore, it is essential to consider the full stochastic dynamics to understand the emergence of hyperuniformity in chemotactic active matter.

Let $V = L^d$ denote the system volume. Under periodic boundary conditions, the allowed wavevectors are $\mathbf{k} = 2\pi\mathbf{n}/L$. We use the symmetric Fourier convention

$$\delta\rho(\mathbf{r},t) = \frac{1}{\sqrt{V}}\sum_{\mathbf{k}}\delta\rho_{\mathbf{k}}(t)e^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}}, \qquad \delta\rho_{\mathbf{k}}(t) = \frac{1}{\sqrt{V}}\int_V \mathrm{d}\mathbf{r}\, e^{-\mathrm{i}\mathbf{k}\cdot\mathbf{r}}\delta\rho(\mathbf{r},t), \tag{45a}$$

$$\delta c(\mathbf{r},t) = \frac{1}{\sqrt{V}}\sum_{\mathbf{k}}\delta c_{\mathbf{k}}(t)e^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}}, \qquad \delta c_{\mathbf{k}}(t) = \frac{1}{\sqrt{V}}\int_V \mathrm{d}\mathbf{r}\, e^{-\mathrm{i}\mathbf{k}\cdot\mathbf{r}}\delta c(\mathbf{r},t). \tag{45b}$$

The factor $1/\sqrt{V}$ is chosen so that the Fourier-space density covariance has the same units as the mean number density. With this convention, the corresponding Fourier component of the white-noise field is

$$\mathbf{\Lambda}_{\mathbf{k}}(t) = \frac{1}{\sqrt{V}}\int_V \mathrm{d}\mathbf{r}\, e^{-\mathrm{i}\mathbf{k}\cdot\mathbf{r}}\mathbf{\Lambda}(\mathbf{r},t), \tag{46}$$

and

$$\nabla\cdot\left[\sqrt{2D\rho_0}\mathbf{\Lambda}(\mathbf{r},t)\right] = \frac{1}{\sqrt{V}}\sum_{\mathbf{k}}\mathrm{i}\mathbf{k}\cdot\sqrt{2D\rho_0}\mathbf{\Lambda}_{\mathbf{k}}(t)e^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}}. \tag{47}$$

After the Fourier transform, the linearized equations become:

$$\frac{\partial}{\partial t}\delta\rho_{\mathbf{k}} = -Dk^2\delta\rho_{\mathbf{k}} + \alpha\rho_0 k^2\delta c_{\mathbf{k}} + \eta_{\mathbf{k}} , \tag{48a}$$

$$\frac{\partial}{\partial t}\delta c_{\mathbf{k}} = -(D_c k^2 - \mu)\delta c_{\mathbf{k}} - \delta\rho_{\mathbf{k}} , \tag{48b}$$

where $\eta_{\mathbf{k}} = -\mathrm{i}\mathbf{k}\cdot\sqrt{2D\rho_0}\mathbf{\Lambda}_{\mathbf{k}}$. The noise has zero mean and covariance

$$\langle\eta_{\mathbf{k}}(t)\eta_{\mathbf{k}'}(t')\rangle = 2D\rho_0 k^2\,\delta_{\mathbf{k},-\mathbf{k}'}\delta(t-t'). \tag{49}$$

Thus, for the conjugate pair $\mathbf{k}' = -\mathbf{k}$, the noise strength entering the covariance of a single mode is $2D\rho_0 k^2$. In a continuous Fourier convention, the Kronecker delta is replaced by the corresponding Dirac delta in wavevector space. Eqs. (48) can be rewritten in matrix form as

$$\partial_t\begin{bmatrix}\delta\rho_{\mathbf{k}}\\ \delta c_{\mathbf{k}}\end{bmatrix} = A(k)\begin{bmatrix}\delta\rho_{\mathbf{k}}\\ \delta c_{\mathbf{k}}\end{bmatrix} + \begin{bmatrix}\eta_{\mathbf{k}}\\ 0\end{bmatrix} , \tag{50}$$

where

$$A(k) = \begin{bmatrix}-Dk^2 & \alpha\rho_0 k^2\\ -1 & \mu - D_c k^2\end{bmatrix} . \tag{51}$$

Let s be the linear growth rate, which is the eigenvalue of $A(k)$. The characteristic equation is given by

$$s^2 - (\mathrm{Tr}A)s + \det A = 0 . \tag{52}$$

where $\mathrm{Tr}A = -Dk^2 + \mu - D_c k^2$ and $\det A = Dk^2(D_c k^2 - \mu) + \alpha\rho_0 k^2$. The homogeneous state is linearly unstable if there exists a wavevector k such that the real part of s is positive. This leads to the stability condition:

$$\mathrm{Tr}A < 0, \quad \det A > 0 , \tag{53}$$

which leads to the following inequalities:

$$\begin{cases} \mu < (D + D_c)k^2 \ , \\ \mu < D_c k^2 + \frac{\alpha\rho_0}{D} \ . \end{cases} \tag{54}$$

Therefore, the homogeneous state is linearly stable if $\mu < \min\{(D + D_c)k^2, D_c k^2 + \frac{\alpha\rho_0}{D}\}$ for all k. In particular, for small k, the stability condition reduces to $\mu < 0 < \frac{\alpha\rho_0}{D}$, which means that the homogeneous state is stable if the chemoattractant decays and the chemotactic sensitivity is positive

$$\mu < 0, \quad \alpha > 0 \ . \tag{55}$$

However, even in the stable regime, fluctuations can still play a significant role in shaping the spatial distribution of particles, which is crucial for understanding the emergence of hyperuniformity.

2.2.1 Density spectrum and normalized structure factor

Before calculating the fluctuation spectrum, it is important to distinguish two closely related quantities. With the Fourier convention in Eq. (45), define the density covariance spectrum

$$C_{\rho\rho}(k) \equiv \langle \delta\rho_{\mathbf{k}} \delta\rho_{-\mathbf{k}} \rangle \, , \tag{56}$$

and the conventional dimensionless structure factor

$$S(k) \equiv \frac{C_{\rho\rho}(k)}{\rho_0}. \tag{57}$$

For a microscopic point configuration and a nonzero allowed wavevector, this definition is equivalent to

$$S(k) = \frac{1}{N} \left\langle \left| \sum_{j=1}^{N} \mathrm{e}^{-\mathrm{i}\mathbf{k}\cdot\mathbf{r}_j} \right|^2 \right\rangle . \tag{58}$$

Consequently, a homogeneous Poisson point process has

$$C^{\mathrm{P}}_{\rho\rho}(k) = \rho_0, \qquad S^{\mathrm{P}}(k) = 1, \qquad \mathbf{k} \neq \mathbf{0}. \tag{59}$$

The strictly zero mode is excluded: in a system with fixed particle number, it is constrained by number conservation, whereas hyperuniformity concerns the limit $k \to 0$ through nonzero wavevectors.

In the limit in which the chemical field relaxes sufficiently rapidly compared with the density mode, we may set $\partial_t \delta c_{\mathbf{k}} \approx 0$ in Eq. (48), obtaining

$$\delta c_{\mathbf{k}} \approx -\frac{\delta\rho_{\mathbf{k}}}{D_c k^2 - \mu}. \tag{60}$$

Substitution into the density equation gives

$$\partial_t \delta\rho_{\mathbf{k}} = -k^2 \left[D + \frac{\alpha\rho_0}{D_c k^2 - \mu} \right] \delta\rho_{\mathbf{k}} + \eta_{\mathbf{k}}. \tag{61}$$

Defining

$$D_{\mathrm{eff}}(k) = D + \frac{\alpha\rho_0}{D_c k^2 - \mu}, \tag{62}$$

we obtain an Ornstein–Uhlenbeck process with relaxation rate $k^2 D_{\mathrm{eff}}(k)$. Using Appendix 4.3, its steady-state density covariance is

$$C^{\mathrm{qs}}_{\rho\rho}(k) = \frac{2D\rho_0 k^2}{2k^2 D_{\mathrm{eff}}(k)} = \frac{D\rho_0}{D + \dfrac{\alpha\rho_0}{D_c k^2 - \mu}}, \tag{63}$$

and therefore the normalized structure factor is

$$\boxed{S_{\rm qs}(k) = \frac{C^{\rm qs}_{\rho\rho}(k)}{\rho_0} = \frac{D}{D + \dfrac{\alpha\rho_0}{D_c k^2 - \mu}}}. \tag{64}$$

The numerator of the covariance calculation represents the strength $2D\rho_0 k^2$ of the conserved Dean noise, while the denominator contains twice the relaxation rate. The normalization in Eq. (57) removes the Poisson density scale. In particular, setting $\alpha = 0$ gives $S(k) = 1$, as required for independent Brownian particles.

For fixed $\mu < 0$, the long-wavelength limit of Eq. (64) is

$$S_0 \equiv \lim_{k\to 0} S_{\rm qs}(k) = \frac{D}{D - \dfrac{\alpha\rho_0}{\mu}} = \frac{D|\mu|}{D|\mu| + \alpha\rho_0}. \tag{65}$$

Hence, in the stable regime $\mu < 0$ and $\alpha > 0$,

$$0 < S_0 < 1. \tag{66}$$

The large-scale fluctuations are suppressed relative to a Poisson process, but the nonzero plateau means that the state is not strictly hyperuniform.

Without the quasi-steady approximation, the full two-component Ornstein–Uhlenbeck system in Eq. (50) can be solved using the Lyapunov equation. As derived in Appendix 4.4, the normalized structure factor is

$$\boxed{S_{\rm full}(k) = D\,\frac{\alpha\rho_0 k^2 + (D_c k^2 - \mu)\left[(D + D_c)k^2 - \mu\right]}{\left[(D + D_c)k^2 - \mu\right]\left[D(D_c k^2 - \mu) + \alpha\rho_0\right]}}. \tag{67}$$

This expression satisfies the consistency checks $S_{\rm full}(k) = 1$ when $\alpha = 0$ and $S_{\rm full}(k) \to 1$ as $k \to \infty$. For any fixed $\mu < 0$, its $k \to 0$ limit is exactly Eq. (65). The point $\mu = 0$, however, is singular and must be treated separately because the chemical relaxation rate $D_c k^2 - \mu$ then also vanishes as $k \to 0$.

2.2.2 Hyperuniformity and the order of limits

A statistically homogeneous point pattern is hyperuniform when its conventional structure factor satisfies

$$\lim_{k\to 0} S(k) = 0. \tag{68}$$

For every fixed $\mu < 0$ and finite $D > 0$, Eq. (65) gives a positive long-wavelength plateau. Thus the present linear theory predicts suppressed, but not strictly hyperuniform, density fluctuations in that parameter regime.

Two singular limits require special care. First, $D \to 0$ removes both ordinary particle diffusion and the Dean-noise amplitude. The resulting disappearance of steady-state fluctuations is therefore a trivial zero-noise limit rather than a nontrivial fluctuation-suppression mechanism.

Second, if one imposes a quasi-static chemical field and then sets $\mu = 0$, Eq. (64) becomes

$$S_{\rm qs}(k) = \frac{1}{1 + \frac{\alpha\rho_0}{DD_c k^2}} \sim \frac{DD_c}{\alpha\rho_0} k^2. \tag{69}$$

Within this instantaneous, elliptic-field approximation, the model displays class-I hyperuniform scaling. However, this conclusion cannot be transferred directly to the full dynamical chemical field. At $\mu = 0$, the chemical relaxation rate is $D_c k^2$, which vanishes at long wavelength; therefore the chemical field is not a fast variable in the limit $k \to 0$, and the adiabatic elimination used above ceases to be uniformly valid.

Setting $\mu = 0$ directly in the full result, Eq. (67), gives

$$S_{\rm full}(k; \mu = 0) = \frac{D}{D + D_c}\,\frac{\alpha\rho_0 + D_c(D + D_c)k^2}{\alpha\rho_0 + DD_c k^2}, \tag{70}$$

so that

$$\lim_{k\to 0} S_{\rm full}(k;\mu=0) = \frac{D}{D+D_c} > 0. \tag{71}$$

The limits $k \to 0$ and $\mu \to 0^-$ therefore do not commute:

$$\lim_{\mu\to 0^-}\lim_{k\to 0} S_{\rm full}(k) = 0, \qquad \lim_{k\to 0}\lim_{\mu\to 0^-} S_{\rm full}(k) = \frac{D}{D+D_c}. \tag{72}$$

For small but nonzero $|\mu|$, a natural crossover wave number is

$$k_\mu \sim \sqrt{\frac{|\mu|}{D_c}}. \tag{73}$$

At $k \ll k_\mu$, the structure factor approaches the finite plateau in Eq. (65). Over a finite intermediate range above this crossover, the spectrum may show an apparent k^2-like suppression if the chemical field is sufficiently fast, but this should be described as finite-scale or "hyperuniform-like" behavior rather than strict asymptotic hyperuniformity.

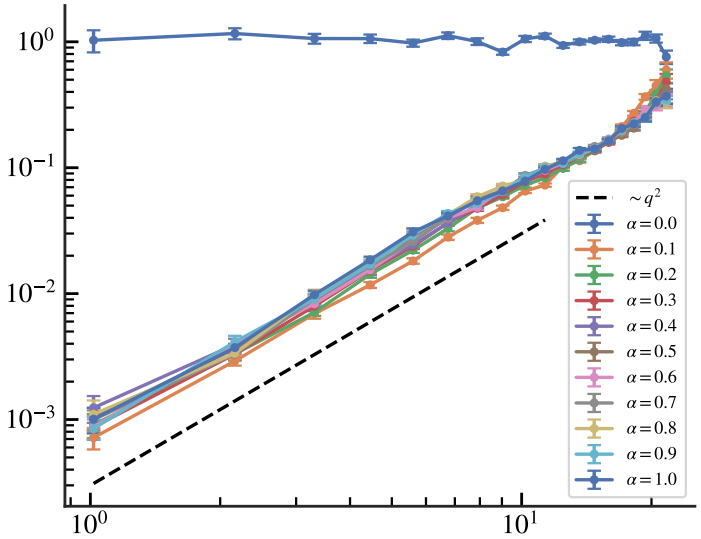


Figure 2: Structure factor as a function of wave number k for different values of α.

3 Example: Feifei's quorum sensing model

3.1 Particle model, sensing kernel, and stochastic convention

Consider particles in two dimensions obeying the Itô equations

$$\mathrm{d}\mathbf{r}_i = \sqrt{2D(P_i)}\,\mathrm{d}\mathbf{W}_i\,, \qquad P_i = \sum_{\substack{j\neq i \\ 0<r_{ij}<d_c}} \frac{1}{2\pi r_{ij}}\,. \tag{74}$$

The Wiener processes are independent. This convention corresponds to evaluating perception before each Euler–Maruyama displacement, as in the continuum construction of Ref. [4], Supplemental Material, Secs. S1–S2. Self-sensing must be excluded for the singular $1/r$ kernel. A microscopic regularization at distances $r \lesssim a$ can be included; all statements below concern wavelengths much larger than a. Unless specified otherwise, the sensing region is a full isotropic disk. The disk formulas apply directly on a periodic square when $d_c < L/2$; a system-spanning minimum-image kernel requires its actual periodic Fourier coefficients.

The diffusivity used in the simulations is the increasing threshold law

$$D(P) = \begin{cases} D_{\min}, & P \le P_{\mathrm{th}}, \\ D_{\max}, & P > P_{\mathrm{th}}, \end{cases} \qquad 0 < D_{\min} < D_{\max}. \tag{75}$$

Particles in regions of larger perceived crowding become more mobile. The analysis below approximates this discontinuous response by a smooth increasing function, while keeping the nonlocal sensing kernel and the conserved noise. The smoothing width is an effective constitutive parameter, not a parameter specified by the microscopic step itself.

Introduce the unnormalized sensing kernel and its continuum field,

$$K_{d_c}(r) = \frac{\Theta(d_c - r)}{2\pi r}, \qquad P(\mathbf{r}) = \int K_{d_c}(|\mathbf{r} - \mathbf{r}'|)\rho(\mathbf{r}')\,\mathrm{d}^2 r'. \tag{76}$$

At the microscopic level the convolution is understood with the self contribution removed, or with a regularized kernel and an explicit constant self subtraction. At the smooth-density level,

$$\int K_{d_c}(r)\,\mathrm{d}^2 r = d_c, \qquad \bar{P} = \rho_0 d_c. \tag{77}$$

Microscopic self subtraction and short-distance regularization can shift this background and regular coefficients without changing the long-distance $1/r$ tail. Unlike the normalized disk kernel in Ref. [4], this kernel has integral d_c, not unity. Dividing it by d_c changes the response variable and requires transforming $D(P)$ accordingly.

3.2 Stochastic continuum equation and linear-noise approximation

For a smooth test function f, Itô's formula gives

$$\mathrm{d}\sum_i f(\mathbf{r}_i) = \sum_i \left[D(P_i)\nabla^2 f(\mathbf{r}_i)\,\mathrm{d}t + \sqrt{2D(P_i)}\nabla f(\mathbf{r}_i)\cdot \mathrm{d}\mathbf{W}_i \right]. \tag{78}$$

Integration by parts and matching the random-current covariance yield the formal Dean equation

$$\partial_t \rho = \nabla^2[\rho D(P)] - \nabla\cdot\left[\sqrt{2\rho D(P)}\,\mathbf{\Lambda}\right], \qquad P = K_{d_c}(r) * \rho(\mathbf{r}, t)\ , \tag{79}$$

where $*$ denotes convolution. The vector white noise has the unit covariance used in Eq. (49) before Fourier transformation. The microscopic identity requires the above self convention; using smooth fields and closing their constitutive response is a coarse-graining approximation.

First assume that $D(P)$ is differentiable near $\bar{P}$ and define

$$\bar{D} = D(\bar{P}) > 0, \qquad g = D'(\bar{P}), \qquad \rho = \rho_0 + \delta\rho, \qquad \delta P = K_{d_c} * \delta\rho. \tag{80}$$

The leading fluctuation expansion of Sec. 1 gives

$$\partial_t \delta\rho = \nabla^2[\bar{D}\delta\rho + \rho_0 g(K_{d_c} * \delta\rho)] - \nabla\cdot[\sqrt{2\rho_0\bar{D}}\,\mathbf{\Lambda}]. \tag{81}$$

Using the volume-normalized Fourier convention of Eq. (45), every nonzero mode satisfies

$$\partial_t \delta\rho_{\mathbf{k}} = -\lambda(k)\delta\rho_{\mathbf{k}} + \eta_{\mathbf{k}}, \tag{82}$$

$$\lambda(k) = k^2[\bar{D} + \rho_0 g\widehat{K}_{d_c}(k)], \tag{83}$$

$$\langle \eta_{\mathbf{k}}(t)\eta_{\mathbf{k}'}(t')\rangle = 2\rho_0\bar{D}k^2\delta_{\mathbf{k},-\mathbf{k}'}\delta(t-t')\ , \tag{84}$$

where $\widehat{K}_{d_c}(k) = \int_{\mathbb{R}^2} K_{d_c}(|\mathbf{r}|)\,\mathrm{e}^{-\mathrm{i}\mathbf{k}\cdot\mathbf{r}}\mathrm{d}^2 r$ is the Fourier transform of the sensing kernel. Thus the scalar version of the Lyapunov equation is

$$-2\lambda(k)C_{\rho\rho}(k) + 2\rho_0\bar{D}k^2 = 0. \tag{85}$$

Whenever $\bar{D} + \rho_0 g\widehat{K}_{d_c}(k) > 0$, the stationary structure factor is

$$\boxed{S(k) = \frac{C_{\rho\rho}(k)}{\rho_0} = \frac{\bar{D}}{\bar{D} + \rho_0 g\widehat{K}_{d_c}(k)}.} \tag{86}$$

This is exact for the stated linear Gaussian closure, not for the nonlinear particle process. It gives $S = 1$ for constant diffusivity.

3.3 Fourier kernel and the origin of an approximately linear spectrum

Sensing kernel $K_{d_c}(r)$ is isotropic, so its Fourier transform depends only on the magnitude $k = |\mathbf{k}|$. In two dimensions, the Fourier transform of an isotropic function can be expressed using the zeroth-order Bessel function of the first kind, $J_0(kr)$:

$$\begin{aligned}\widehat{K}_{d_c}(k) &= \int_0^{d_c}\int_0^{2\pi} \frac{1}{2\pi r} \mathrm{e}^{-\mathrm{i}kr\cos\theta} r \mathrm{d}\theta \mathrm{d}r \\ &= \int_0^{d_c} J_0(kr)\,\mathrm{d}r \qquad\qquad F(x) = \int_0^x J_0(u)\,\mathrm{d}u. \\ &= \frac{F(kd_c)}{k}\,,\end{aligned} \tag{87}$$

At small argument, one can expand $J_0(u) = 1 - u^2/4 + u^4/64 + O(u^6)$ and integrate term by term to obtain

$$\widehat{K}_{d_c}(k) = d_c\left[1 - \frac{(kd_c)^2}{12} + \frac{(kd_c)^4}{320} + O((kd_c)^6)\right]. \tag{88}$$

At large argument, $F(x) = 1 + O(x^{-1/2})$, with an oscillatory correction from the sharp sensing boundary (For details, see Appendix 4.5). Consequently,

$$\widehat{K}_{d_c}(k) = \frac{1}{k}[1 + O((kd_c)^{-1/2})], \qquad d_c^{-1} \ll k \ll a^{-1}. \tag{89}$$

The upper restriction is relevant when a microscopic cutoff a is introduced. It also must lie within the continuum regime.

For an increasing response $g > 0$, define the inverse length and dimensionless feedback strength

$$k_* \equiv \frac{\rho_0 g}{\bar{D}}, \qquad B \equiv k_* d_c. \tag{90}$$

Equations (86) and (87) give the complete finite-range prediction

$$\boxed{S(k) = \left[1 + \frac{k_*}{k} F(kd_c)\right]^{-1}}. \tag{91}$$

If $B \gg 1$, there is a separated interval

$$\boxed{d_c^{-1} \ll k \ll k_*, \qquad S(k) \simeq \frac{k}{k_*} = \frac{\bar{D}}{\rho_0 g}\, k.} \tag{92}$$

One must also impose the microscopic and continuum upper cutoffs. In this interval the relaxation rate is $\lambda \simeq \rho_0 g k$, whereas the conserved noise power remains proportional to k^2. Their ratio gives the exponent one.

Fixed finite sensing radius. For any fixed finite d_c and finite regular coefficients, the actual $k \to 0$ prediction is

$$S_0 = \frac{1}{1+B} > 0, \tag{93}$$

$$S(k) = \frac{1}{1+B}\left[1 + \frac{B}{12(1+B)}(kd_c)^2 + O((kd_c)^4)\right]. \tag{94}$$

Thus the linear closure predicts a suppressed plateau followed, when $B \gg 1$, by a k-like interval, and eventually a return toward $S = 1$. The local logarithmic slope approaches zero at the smallest k, even if it is close to one over a broad accessible interval. The observation of $S \sim k$ at finite size is therefore consistent with this mechanism without establishing asymptotic hyperuniformity. This conclusion concerns the homogeneous linear theory; it is not a no-go theorem for nonlinear or absorbing-state mechanisms.

3.4 Continuous approximations to the increasing threshold response

The literal step in Eq. (75) is not differentiable at threshold. At a homogeneous background strictly on either branch, its ordinary derivative is zero, so a naive infinitesimal linearization predicts $S = 1$. Exactly at the threshold no such Taylor expansion exists. Assigning an arbitrary positive number to $D'(\bar{P})$ is therefore not a derivation for the step model.

One controlled modification is a prescribed smooth switching law, for example

$$D_\epsilon(P) = D_{\min} + \frac{D_{\max} - D_{\min}}{1 + \exp[-(P - P_{\rm th})/\epsilon]}. \tag{95}$$

Equivalently, this is a hyperbolic-tangent response with argument $(P - P_{\rm th})/(2\epsilon)$. Writing

$$q = \frac{1}{1 + \exp[-(\bar{P} - P_{\rm th})/\epsilon]}, \qquad \Delta D = D_{\max} - D_{\min}, \tag{96}$$

its operating-point (the fixed point where the system is linearized) coefficients are

$$\bar{D} = D_{\min} + \Delta D q, \qquad g = \frac{\Delta D}{\epsilon} q(1-q), \qquad k_* = \frac{\rho_0 \Delta D q(1-q)}{\epsilon(D_{\min} + \Delta D q)}. \tag{97}$$

These expressions quantify how moving away from the switching region reduces the response slope and eventually eliminates the linear scaling window. The linear-noise expansion requires perception fluctuations small compared with the response width ϵ; taking $\epsilon \to 0$ at fixed fluctuations is not controlled.

For a literal threshold process, unresolved perception fluctuations may smooth the *coarse-grained* response. As a stated closure, suppose the perception of a tagged particle is $P = m + \zeta$, with a normalized, smooth probability density p_ζ whose shape is unchanged by a weak slow modulation of m. Then

$$D_{\rm eff}(m) = D_{\min} + (D_{\max} - D_{\min}) \int_{P_{\rm th} - m}^{\infty} p_\zeta(z)\, {\rm d}z, \tag{98}$$

$$g_{\rm eff}(m) = (D_{\max} - D_{\min}) p_\zeta(P_{\rm th} - m) \geq 0. \tag{99}$$

If these fast fluctuations can be eliminated into a local instantaneous constitutive law with noise mobility $\rho_0 \bar{D}_{\rm eff}$, Eq. (91) follows with $k_* = \rho_0 g_{\rm eff}/\bar{D}_{\rm eff}$. This is a testable effective closure, not an exact consequence of the microscopic step. Changes in the conditional perception distribution, slow switching, and density–mobility correlations can change both the drift and effective noise. In particular, the width of p_ζ cannot be inferred from the threshold parameters alone. The singular kernel makes short-distance statistics relevant to that width; a microscopic cutoff may be needed for finite perception variance.

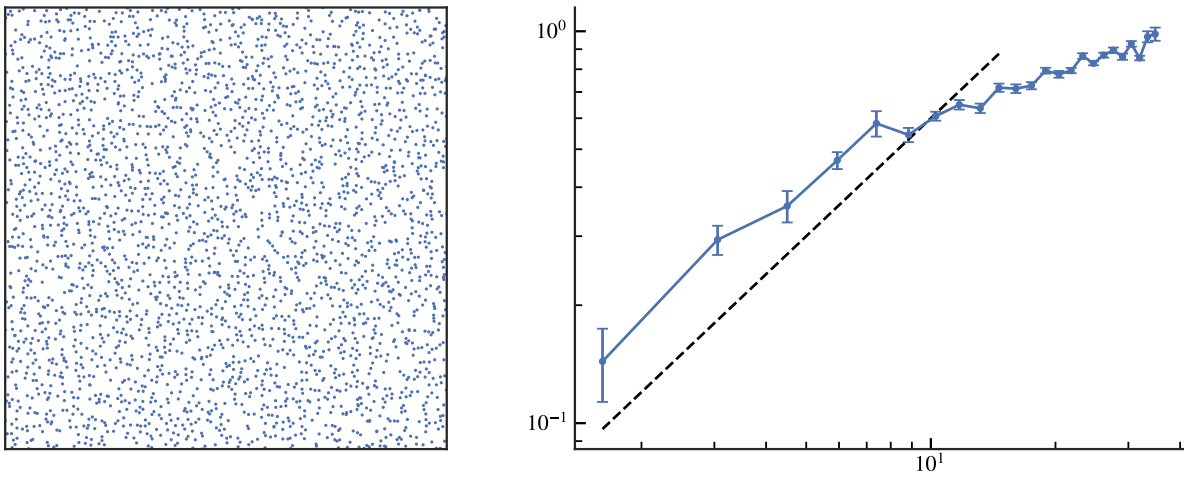


Figure 3: Snapshot and structure factor of the simulated Feifei's quorum sensing model. (a) A representative configuration of $N = 1024$ particles in a periodic square with $L = 32$, $d_c = 1.6$, $D_{\min} = 0.01$, $D_{\max} = 0.1$, and $P_{\rm th} = 0.5$. (b) The corresponding structure factor $S(k)$ as a function of wave number k. The dashed line indicates the predicted linear scaling regime, while the solid line represents the full theory from Eq. (91).

4 Appendix

4.1 Derivation of the linear-noise approximation

This appendix derives Eq. (4) from Eq. (1). The essential point is that the physical fluctuation and the imposed noise amplitude are both of order $\sqrt{\varepsilon}$. The dependence of the noise amplitude on that fluctuation therefore first enters at order ε in the original field equation.

4.1.1 A route using fixed-point linearization and ordinary Taylor series

The derivation can be understood using the same steps as a stability calculation near a fixed point. We first keep a fixed spatial grid, as already assumed by the spatial cutoff in the main text. This changes the notation to ordinary vectors and matrices without introducing an additional approximation to that regularized model. We do not assert that taking the grid spacing to zero is automatically justified. The detailed operator notation and stochastic error estimate are supplied in the following subsubsection.

Start with the familiar deterministic calculation. For $\dot{u} = f(u)$ and a fixed point $f(u_0) = 0$, substitute $u = u_0 + \delta u$ and expand:

$$\dot{\delta u} = f'(u_0)\delta u + \frac{1}{2} f''(u_0)(\delta u)^2 + \cdots . \tag{100}$$

The first term describes the leading restoring or destabilizing motion. The stochastic calculation uses this same Taylor expansion, but must also expand the amplitude of the random forcing.

Why use $\sqrt{\varepsilon}$ for the fluctuation size? Write $a = \sqrt{\varepsilon}$ to make the powers easier to follow. Here a controls the noise amplitude and $a^2 = \varepsilon$ controls its variance. For example, the stable scalar equation

$$\mathrm{d}\delta u = -\kappa \delta u \,\mathrm{d}t + a g_0 \,\mathrm{d}W_t, \qquad \kappa > 0, \tag{101}$$

has stationary variance and root-mean-square fluctuation

$$\langle (\delta u)^2 \rangle = \frac{a^2 g_0^2}{2\kappa}, \qquad \sqrt{\langle (\delta u)^2 \rangle} = \frac{a|g_0|}{\sqrt{2\kappa}}. \tag{102}$$

At fixed κ, the fluctuation is therefore of order a. Writing $\delta u = av$ measures the small physical fluctuation in units of its expected size: v stays of order one as $a \to 0$. This is a motivated scaling assumption, not a universal conclusion for every stochastic system. Near a vanishing restoring rate, or with a large initial perturbation, it requires a separate check. In particular, a small-noise limit at fixed nonzero k need not be uniform as $k \to 0$.

Derive the result for one variable. Consider

$$\mathrm{d}u = f(u)\,\mathrm{d}t + a g(u)\,\mathrm{d}W_t, \qquad f(u_0) = 0, \qquad u = u_0 + a v^{\varepsilon}. \tag{103}$$

The superscript only records that the exact rescaled fluctuation can still depend on ε. Ordinary Taylor series give

$$f(u_0 + a v^{\varepsilon}) = a f'(u_0) v^{\varepsilon} + \frac{a^2}{2} f''(u_0)(v^{\varepsilon})^2 + O(a^3), \tag{104}$$

$$a g(u_0 + a v^{\varepsilon}) = a g(u_0) + a^2 g'(u_0) v^{\varepsilon} + O(a^3). \tag{105}$$

Since $\mathrm{d}u = a\,\mathrm{d}v^{\varepsilon}$, divide both sides by a:

$$\begin{aligned} \mathrm{d}v^{\varepsilon} &= f'(u_0) v^{\varepsilon}\,\mathrm{d}t + g(u_0)\,\mathrm{d}W_t \\ &\quad + a\left[\frac{1}{2} f''(u_0)(v^{\varepsilon})^2\,\mathrm{d}t + g'(u_0) v^{\varepsilon}\,\mathrm{d}W_t\right] + O(a^2)\,\mathrm{d}t + O(a^2)\,\mathrm{d}W_t. \end{aligned} \tag{106}$$

There is now a direct separation: the first line has no small prefactor, while the second line has the small prefactor a. Keeping the leading equation gives

$$\boxed{\mathrm{d}v = f'(u_0)v\,\mathrm{d}t + g(u_0)\,\mathrm{d}W_t.} \tag{107}$$

Thus the random forcing itself remains. Only the change in its amplitude caused by the small displacement from u_0 is higher order. No averaging of the noise to zero has been used.

Several variables: replace $f'(u_0)$ by the Jacobian. On a fixed grid of M sites, stack all n fields at all sites into a vector U with $N = nM$ entries. Use P independent Brownian noise channels, absorbing the grid-volume normalization into their coefficient matrix G. The Itô equations are ordinary coupled stochastic equations,

$$\mathrm{d}U_i = F_i(U)\,\mathrm{d}t + a\sum_{b=1}^{P} G_{ib}(U)\,\mathrm{d}W_b, \qquad i = 1,\ldots,N. \tag{108}$$

For a fixed point $F(U_0) = 0$, let $U = U_0 + aV^\varepsilon$. Define the usual Jacobian

$$J_{ij} = \left.\frac{\partial F_i}{\partial U_j}\right|_{U_0}. \tag{109}$$

Expanding every component gives

$$F_i(U_0 + aV^\varepsilon) = a\sum_j J_{ij}V_j^\varepsilon + \frac{a^2}{2}\sum_{j,\ell}\left.\frac{\partial^2 F_i}{\partial U_j \partial U_\ell}\right|_{U_0} V_j^\varepsilon V_\ell^\varepsilon + O(a^3), \tag{110}$$

$$aG_{ib}(U_0 + aV^\varepsilon) = aG_{ib}(U_0) + a^2\sum_j \left.\frac{\partial G_{ib}}{\partial U_j}\right|_{U_0} V_j^\varepsilon + O(a^3). \tag{111}$$

Division by a and retention of the leading terms produce

$$\mathrm{d}V_i = \sum_j J_{ij}V_j\,\mathrm{d}t + \sum_b G_{ib}(U_0)\,\mathrm{d}W_b. \tag{112}$$

In field notation, the Jacobian action JV is written $\mathcal{L}\mathbf{v}$, and the matrix acting on the noise is written $\mathbf{G}[\mathbf{u}_0]$. Equation (112) is exactly the grid version of

$$\boxed{\partial_t \mathbf{v} = \mathcal{L}\mathbf{v} + \mathbf{G}[\mathbf{u}_0]\boldsymbol{\xi}.} \tag{113}$$

The substantive approximation is the stated leading-order expansion in noise strength; writing its finite-grid Jacobian in operator notation does not introduce a new physical assumption.

What is a Fréchet derivative? This term comes from functional analysis, the extension of vector-space calculus to spaces whose elements can be whole functions. It expresses the same local linear approximation as a Jacobian. More precisely, for chosen input and output norms, a linear map $\mathcal{L}$ is the Fréchet derivative when

$$\frac{\|\mathbf{F}[\mathbf{u}_0 + \mathbf{h}] - \mathbf{F}[\mathbf{u}_0] - \mathcal{L}\mathbf{h}\|}{\|\mathbf{h}\|} \longrightarrow 0 \quad \text{as } \|\mathbf{h}\| \longrightarrow 0. \tag{114}$$

The condition says that the error of the linear approximation becomes small compared with the perturbation itself. At fixed grid size this is precisely ordinary differentiability with $\mathcal{L} = J$. For an unregularized PDE, the spaces and norms must be chosen to handle spatial derivatives; this extra mathematical issue is not being silently assumed away here.

For example, for the scalar reaction–diffusion drift $F[u] = D\nabla^2 u + ru - bu^3$ and a uniform fixed point u_0, direct substitution gives

$$F[u_0 + h] - F[u_0] = \underbrace{D\nabla^2 h + (r - 3bu_0^2)h}_{\mathcal{L}h} - 3bu_0 h^2 - bh^3. \tag{115}$$

One can simply collect the terms linear in h to obtain $\mathcal{L}$. The name "Fréchet derivative" provides a general definition of that operation; it is not an additional calculation required for this example.

Why does white noise not invalidate the power counting? Brownian motion has independent increments with $\mathbb{E}\Delta W_b = 0$ and $\mathbb{E}(\Delta W_b \Delta W_c) = \delta_{bc}\Delta t$. The notation $\xi = \dot{W}$ is shorthand: it has no finite ordinary value at a single instant. To check smallness, compare accumulated random increments rather than pointwise values of ξ.

Let $H_{ib}(t)$ be the coefficient of a discarded noise term in the rescaled equation. During one short step, its contribution has the form

$$\Delta Z_i = a \sum_b H_{ib}(t) \Delta W_b. \tag{116}$$

The Itô convention uses the coefficient evaluated from the state at the beginning of the step. Conditional on that state, the new Brownian increments are independent of it, so

$$\mathbb{E}\left[\sum_i (\Delta Z_i)^2 \middle| \text{state at } t \right] = a^2 \Delta t \sum_{i,b} H_{ib}(t)^2. \tag{117}$$

For such left-endpoint sums, cross terms between distinct time steps vanish by conditioning on the earlier history. Passing to the stochastic integral gives the exact identity

$$\mathbb{E} \sum_i \left(a \sum_b \int_0^T H_{ib}(s)\, \mathrm{d}W_b(s) \right)^2 = a^2 \int_0^T \mathbb{E} \sum_{i,b} H_{ib}(s)^2 \, \mathrm{d}s. \tag{118}$$

This is the finite-dimensional Itô isometry. If the integral on the right stays bounded as $a \to 0$, the root-mean-square size of the discarded accumulated noise is $O(a)$. This is why the extra power of a really makes the correction small. It is an estimate of the discarded forcing; controlling the resulting trajectory error additionally uses smooth drift coefficients and bounded fluctuations on the time interval considered.

What are the Frobenius and Hilbert–Schmidt norms? The Frobenius norm is a matrix norm from linear algebra. It is simply the Euclidean length obtained by listing all the matrix entries:

$$\|H\|_{\mathrm{F}} = \left(\sum_{i,b} |H_{ib}|^2 \right)^{1/2}. \tag{119}$$

For instance, if $H = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$, then $\|H\|_{\mathrm{F}}^2 = 1+4+0+9 = 14$ and $\|H\|_{\mathrm{F}} = \sqrt{14}$. It is the *squared* norm, not the norm itself, that sums the squared amplitudes in Eq. (118).

The Hilbert–Schmidt norm comes from operator theory and functional analysis. It extends this idea to a linear operator H between Hilbert spaces, which are vector spaces equipped with an inner product and the corresponding notion of length. For an orthonormal basis $\{e_b\}$ of the input noise space,

$$\|H\|_{\mathrm{HS}}^2 = \sum_b \|H e_b\|^2. \tag{120}$$

It adds the squared output produced by each independent unit input. For a finite matrix in orthonormal coordinates it equals the Frobenius norm. With infinitely many channels the sum must converge; spatial white noise and differential noise operators can fail this requirement without regularization. This is why the fixed cutoff matters. For the finite-grid derivation, every occurrence of $\|H\|_{\mathrm{HS}}^2$ can be read literally as $\sum_{i,b} |H_{ib}|^2$.

What has and has not been approximated

The Jacobian calculation and the component-wise noise-variance identity are ordinary finite-dimensional versions of the operator statements below. They retain the same rigor at the fixed spatial cutoff. The approximation is to retain the leading small-noise equation, under smooth coefficients, initial fluctuations of order $\sqrt{\varepsilon}$, and bounded rescaled fluctuations over the time

interval considered. Neither the continuum limit nor a uniform stationary or long-wavelength error bound is proved by this local expansion. These qualifications are retained throughout.

4.1.2 Operator form, explicit remainder terms, and the chemotaxis example

The following derivation records the same calculation in compact field notation. In it, $\mathcal{L}$ is the field version of the Jacobian, $\mathbf{F}_0^{(2)}[h,h]$ is the component-wise Hessian sum, and $\mathbf{G}_0^{(1)}[h]$ means differentiating each entry of the noise matrix in the direction h.

1. Stochastic equation and expansion assumptions. To avoid differentiating white noise as an ordinary function, write the Itô equation in differential form,

$$\mathrm{d}\mathbf{u}^\varepsilon = \mathbf{F}[\mathbf{u}^\varepsilon]\,\mathrm{d}t + \sqrt{\varepsilon}\,\mathbf{G}[\mathbf{u}^\varepsilon]\,\mathrm{d}\mathbf{W}_t, \tag{121}$$

where $\boldsymbol{\xi} = \mathrm{d}\mathbf{W}_t/\mathrm{d}t$ is only formal notation. Work at a fixed spatial regularization, so the stochastic integrals and derivatives of $\mathbf{F}$ and $\mathbf{G}$ are defined. Assume these maps are sufficiently smooth near a deterministic stationary state $\mathbf{u}_0$, with $\mathbf{F}[\mathbf{u}_0] = 0$. Take initial data $\mathbf{u}^\varepsilon(0) = \mathbf{u}_0 + \sqrt{\varepsilon}\,\mathbf{v}^\varepsilon(0)$ with scaled fluctuations bounded in the moments needed below. The expansion applies on time intervals where those fluctuations remain bounded; uniformity as $t \to \infty$ or $k \to 0$ needs additional estimates.

Define the exact rescaled fluctuation at each $\varepsilon > 0$ by

$$\mathbf{v}^\varepsilon = \frac{\mathbf{u}^\varepsilon - \mathbf{u}_0}{\sqrt{\varepsilon}}. \tag{122}$$

Since $\mathbf{u}_0$ is constant in time and this change of variables is affine, Itô's formula gives $\mathrm{d}\mathbf{u}^\varepsilon = \sqrt{\varepsilon}\,\mathrm{d}\mathbf{v}^\varepsilon$ with no second-derivative correction. Equation (121) therefore becomes exactly

$$\mathrm{d}\mathbf{v}^\varepsilon = \frac{\mathbf{F}[\mathbf{u}_0 + \sqrt{\varepsilon}\,\mathbf{v}^\varepsilon] - \mathbf{F}[\mathbf{u}_0]}{\sqrt{\varepsilon}}\,\mathrm{d}t + \mathbf{G}[\mathbf{u}_0 + \sqrt{\varepsilon}\,\mathbf{v}^\varepsilon]\,\mathrm{d}\mathbf{W}_t. \tag{123}$$

2. Functional Taylor expansion of the drift. Let $D\mathbf{F}[\mathbf{u}_0]$ denote the first Fréchet derivative, acting on a perturbation field $\mathbf{h}$ as

$$(\mathcal{L}\mathbf{h})_a(\mathbf{r}) = \sum_{b=1}^{n} \int_V \mathrm{d}\mathbf{r}' \left.\frac{\delta F_a[\mathbf{u}](\mathbf{r})}{\delta u_b(\mathbf{r}')}\right|_{\mathbf{u}_0} h_b(\mathbf{r}'), \qquad \mathcal{L} = D\mathbf{F}[\mathbf{u}_0]. \tag{124}$$

For local differential equations this kernel can contain derivatives of delta functions; its action is the usual linear differential operator. Write $\mathbf{F}_0^{(2)} = D^2\mathbf{F}[\mathbf{u}_0]$ for the bilinear second derivative. For a bounded perturbation $\mathbf{h}$,

$$\mathbf{F}[\mathbf{u}_0 + \sqrt{\varepsilon}\,\mathbf{h}] = \underbrace{\mathbf{F}[\mathbf{u}_0]}_{=0} + \sqrt{\varepsilon}\,\mathcal{L}\mathbf{h} + \frac{\varepsilon}{2}\mathbf{F}_0^{(2)}[\mathbf{h},\mathbf{h}] + O(\varepsilon^{3/2}). \tag{125}$$

The first nonzero drift term is consequently of order $\sqrt{\varepsilon}$ in the physical equation, and of order one after dividing by $\sqrt{\varepsilon}$.

3. Expansion of the multiplicative noise coefficient. Set $\mathbf{G}_0 = \mathbf{G}[\mathbf{u}_0]$ and $\mathbf{G}_0^{(1)} = D\mathbf{G}[\mathbf{u}_0]$. Its Taylor expansion is

$$\mathbf{G}[\mathbf{u}_0 + \sqrt{\varepsilon}\,\mathbf{h}] = \mathbf{G}_0 + \sqrt{\varepsilon}\,\mathbf{G}_0^{(1)}[\mathbf{h}] + O(\varepsilon). \tag{126}$$

Here $\mathbf{G}_0^{(1)}[\mathbf{h}]$ is still an operator mapping noise channels into field increments. Including the explicit prefactor in Eq. (121) gives

$$\sqrt{\varepsilon}\,\mathbf{G}[\mathbf{u}_0 + \sqrt{\varepsilon}\,\mathbf{h}]\,\mathrm{d}\mathbf{W}_t = \sqrt{\varepsilon}\,\mathbf{G}_0\,\mathrm{d}\mathbf{W}_t + \varepsilon\,\mathbf{G}_0^{(1)}[\mathbf{h}]\,\mathrm{d}\mathbf{W}_t + O(\varepsilon^{3/2})\,\mathrm{d}\mathbf{W}_t. \tag{127}$$

The powers here count ε, not the time increment. For example, Itô isometry at fixed regularization gives

$$\mathbb{E}\left\|\sqrt{\varepsilon}\int_0^T H_s\,\mathrm{d}\mathbf{W}_s\right\|^2 = \varepsilon\,\mathbb{E}\int_0^T \|H_s\|_{\mathrm{HS}}^2\,\mathrm{d}s, \tag{128}$$

where the squared Hilbert–Schmidt norm is the sum of squared noise-channel amplitudes: $\|H\|_{\mathrm{HS}}^2 = \sum_{i,b}|H_{ib}|^2$ for a finite matrix in orthonormal coordinates. This is exactly the component-wise variance calculation in Eq. (118). Thus the stochastic correction is small in mean square when the integral on the right is uniformly bounded. White noise does not cancel the extra factor of $\sqrt{\varepsilon}$.

4. Collecting orders and taking the leading equation. Substituting the expansions into Eq. (123) yields

$$\begin{aligned}\mathrm{d}\mathbf{v}^\varepsilon &= \mathcal{L}\mathbf{v}^\varepsilon\,\mathrm{d}t + \mathbf{G}_0\,\mathrm{d}\mathbf{W}_t \\ &\quad + \sqrt{\varepsilon}\left\{\frac{1}{2}\mathbf{F}_0^{(2)}[\mathbf{v}^\varepsilon,\mathbf{v}^\varepsilon]\,\mathrm{d}t + \mathbf{G}_0^{(1)}[\mathbf{v}^\varepsilon]\,\mathrm{d}\mathbf{W}_t\right\} + O(\varepsilon)\,\mathrm{d}t + O(\varepsilon)\,\mathrm{d}\mathbf{W}_t.\end{aligned} \tag{129}$$

The remainder notation refers to the drift and stochastic-integrand coefficients on bounded sets. Retaining only terms of order one in this rescaled equation defines the leading fluctuation $\mathbf{v}$:

$$\mathrm{d}\mathbf{v} = \mathcal{L}\mathbf{v}\,\mathrm{d}t + \mathbf{G}_0\,\mathrm{d}\mathbf{W}_t, \qquad \boxed{\partial_t\mathbf{v} = \mathcal{L}\mathbf{v} + \mathbf{G}[\mathbf{u}_0]\boldsymbol{\xi}.} \tag{130}$$

In particular, the term $\mathbf{G}_0^{(1)}[\mathbf{v}]\boldsymbol{\xi}$ is not discarded because its mean vanishes. It is discarded because it carries an additional factor $\sqrt{\varepsilon}$ in Eq. (129), at the same order as the quadratic drift correction. At fixed noise strength, a Taylor expansion only in the state perturbation would retain a linear multiplicative-noise term and would not in general give an additive Ornstein–Uhlenbeck process.

For completeness, the formal next-order ansatz $\mathbf{u}^\varepsilon = \mathbf{u}_0 + \sqrt{\varepsilon}\,\mathbf{v} + \varepsilon\mathbf{w} + \cdots$ gives

$$\mathrm{d}\mathbf{w} = \left(\mathcal{L}\mathbf{w} + \frac{1}{2}\mathbf{F}_0^{(2)}[\mathbf{v},\mathbf{v}]\right)\mathrm{d}t + \mathbf{G}_0^{(1)}[\mathbf{v}]\,\mathrm{d}\mathbf{W}_t. \tag{131}$$

This identifies both corrections omitted from the leading equation. With zero-mean leading initial fluctuations, $\mathbb{E}\mathbf{v} = 0$, but the quadratic term can produce an order-ε shift of $\mathbb{E}\mathbf{u}^\varepsilon$ through $\mathbb{E}\mathbf{w}$. The original equation was already Itô; a Stratonovich starting equation would first require its drift conversion, which is of order ε for the noise amplitude in Eq. (121).

5. Physical fluctuation and covariance normalization. The leading physical fluctuation is $\delta\mathbf{u}_{\mathrm{lin}} = \sqrt{\varepsilon}\,\mathbf{v}$, and therefore

$$\partial_t\delta\mathbf{u}_{\mathrm{lin}} = \mathcal{L}\delta\mathbf{u}_{\mathrm{lin}} + \sqrt{\varepsilon}\,\mathbf{G}_0\boldsymbol{\xi}. \tag{132}$$

If $B_0(\mathbf{k})$ is the Fourier noise operator for $\mathbf{v}$, the physical Fourier variable $\mathbf{x}_\mathbf{k}$ in Eq. (6) has

$$B = \sqrt{\varepsilon}\,B_0, \qquad Q = \varepsilon B_0 B_0^\dagger, \qquad C_{\mathrm{lin}} = \varepsilon C_v. \tag{133}$$

Thus removing $\sqrt{\varepsilon}$ from the rescaled equation does not remove the physical noise strength. If ε is only a bookkeeping parameter and is subsequently set to one, the result remains a linear-noise approximation whose accuracy must be assessed for the particular model.

4.2 Derivation of Dean's equation

The continuum equation for the particle density can be derived from the microscopic Langevin dynamics by applying Itô's formula. We start from the microscopic density

$$\rho(\mathbf{r},t) = \sum_{i=1}^N \delta(\mathbf{r} - \mathbf{r}_i(t)). \tag{134}$$

Taking the time derivative and using the chain rule gives the exact continuity equation

$$\partial_t \rho(\mathbf{r}, t) = -\nabla \cdot \mathbf{J}(\mathbf{r}, t), \tag{135}$$

where the microscopic current is

$$\mathbf{J}(\mathbf{r}, t) = \sum_{i=1}^{N} \dot{\mathbf{r}}_i(t)\, \delta(\mathbf{r} - \mathbf{r}_i(t)). \tag{136}$$

Substituting the particle Langevin equation into the current yields two contributions. The chemotactic drift gives

$$\mathbf{J}_{\text{chemo}}(\mathbf{r}, t) = \alpha \sum_{i=1}^{N} \nabla c(\mathbf{r}_i, t)\, \delta(\mathbf{r} - \mathbf{r}_i). \tag{137}$$

Under the local-field approximation $\nabla c(\mathbf{r}_i, t) \approx \nabla c(\mathbf{r}, t)$, this becomes

$$\mathbf{J}_{\text{chemo}}(\mathbf{r}, t) \approx \alpha \rho(\mathbf{r}, t) \nabla c(\mathbf{r}, t), \tag{138}$$

which leads to the deterministic chemotactic term in the density equation.

The Brownian noise contribution requires an Itô expansion. For a single particle,

$$\mathrm{d}\delta(\mathbf{r} - \mathbf{r}_i) = -\nabla \delta(\mathbf{r} - \mathbf{r}_i) \cdot \mathrm{d}\mathbf{r}_i + D \nabla^2 \delta(\mathbf{r} - \mathbf{r}_i)\, \mathrm{d}t. \tag{139}$$

The first term on the right-hand side is the stochastic contribution. Summing over all particles, it can be written as the divergence of a random current:

$$\mathbf{J}_{\text{noise}}(\mathbf{r}, t) = \sqrt{2D} \sum_{i=1}^{N} \delta(\mathbf{r} - \mathbf{r}_i)\, \boldsymbol{\xi}_i(t). \tag{140}$$

The second term gives the diffusive contribution after summation:

$$\sum_{i=1}^{N} D \nabla^2 \delta(\mathbf{r} - \mathbf{r}_i) = D \nabla^2 \rho(\mathbf{r}, t). \tag{141}$$

The random current has the correlation

$$\langle J_{\text{noise},a}(\mathbf{r}, t) J_{\text{noise},b}(\mathbf{r}', t') \rangle = 2D \rho(\mathbf{r}, t) \delta_{ab} \delta(\mathbf{r} - \mathbf{r}') \delta(t - t'). \tag{142}$$

It can therefore be represented by a spatio-temporal white noise field $\boldsymbol{\Lambda}(\mathbf{r}, t)$ with unit variance:

$$\mathbf{J}_{\text{noise}}(\mathbf{r}, t) = \sqrt{2D\rho(\mathbf{r}, t)}\, \boldsymbol{\Lambda}(\mathbf{r}, t), \tag{143}$$

where

$$\langle \Lambda_a(\mathbf{r}, t) \Lambda_b(\mathbf{r}', t') \rangle = \delta_{ab} \delta(\mathbf{r} - \mathbf{r}') \delta(t - t'). \tag{144}$$

Collecting all terms, we arrive at the Dean equation:

$$\boxed{\partial_t \rho = -\alpha \nabla \cdot (\rho \nabla c) + D \nabla^2 \rho - \nabla \cdot \left[\sqrt{2D\rho}\, \boldsymbol{\Lambda} \right].} \tag{145}$$

Under the mean-field approximation, the noise term vanishes on average since $\langle \boldsymbol{\Lambda} \rangle = 0$, giving the Keller-Segel type equation

$$\partial_t \langle \rho \rangle = -\alpha \nabla \cdot (\langle \rho \rangle \nabla c) + D \nabla^2 \langle \rho \rangle. \tag{146}$$

However, this noise term must be retained when calculating fluctuation quantities such as the structure factor $S(k)$.

4.3 Calculation of variance for Ornstein-Uhlenbeck process

For an Ornstein-Uhlenbeck process described by the equation

$$\partial_t x(t) = -\lambda x(t) + \eta(t) \,, \tag{147}$$

where $\eta(t)$ is a Gaussian white noise with zero mean and correlation $\langle \eta(t)\eta(t')\rangle = 2\Gamma\delta(t-t')$, the formal solution is

$$X\left(t\right) = X\left(0\right)\mathrm{e}^{-\lambda t} + \int_{-\infty}^{\top} \mathrm{e}^{-\lambda(t-s)}\eta\left(s\right)\mathrm{d}s \,. \tag{148}$$

After a long time, the initial condition becomes irrelevant:

$$X\left(t\right) \approx \int_{-\infty}^{\top} \mathrm{e}^{-\lambda(t-s)}\eta\left(s\right)\mathrm{d}s \,. \tag{149}$$

The variance of $X(t)$ can be calculated as:

$$\begin{aligned} \langle X^2(t)\rangle &= \int_{-\infty}^{\top}\int_{-\infty}^{\top} \mathrm{e}^{-\lambda(2t-s-s')}\langle\eta(s)\eta(s')\rangle \mathrm{d}s\mathrm{d}s' \\ &= 2\Gamma\int_{-\infty}^{\top} e^{-2\lambda(t-s)}\mathrm{d}s \\ &= \frac{\Gamma}{\lambda} \,. \end{aligned} \tag{150}$$

4.4 Structure factor from the Lyapunov equation

The complete density covariance follows from the steady-state covariance equation of the two-dimensional linear Ornstein–Uhlenbeck system,

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix}\delta\rho_{\mathbf{k}}\\ \delta c_{\mathbf{k}}\end{pmatrix} = A(k)\begin{pmatrix}\delta\rho_{\mathbf{k}}\\ \delta c_{\mathbf{k}}\end{pmatrix} + \begin{pmatrix}\eta_{\mathbf{k}}\\ 0\end{pmatrix}, \tag{151}$$

where

$$A(k) = \begin{pmatrix} -Dk^2 & \alpha\rho_0 k^2 \\ -1 & -(D_c k^2 - \mu)\end{pmatrix}. \tag{152}$$

Introduce

$$a = Dk^2, \qquad b = \alpha\rho_0 k^2, \qquad q = D_c k^2 - \mu, \qquad \Gamma = D\rho_0 k^2, \tag{153}$$

so that

$$A = \begin{pmatrix} -a & b \\ -1 & -q\end{pmatrix}, \qquad Q = \begin{pmatrix} 2\Gamma & 0 \\ 0 & 0\end{pmatrix}. \tag{154}$$

Here Q is the covariance matrix of the additive noise for the conjugate pair $\mathbf{k}$ and $-\mathbf{k}$.

Write the steady covariance matrix as

$$C = \begin{pmatrix} x & y \\ y & z\end{pmatrix}, \tag{155}$$

where

$$x = \langle\delta\rho_{\mathbf{k}}\delta\rho_{-\mathbf{k}}\rangle = C_{\rho\rho}(k) = \rho_0 S(k). \tag{156}$$

The Lyapunov equation

$$AC + CA^{\top} + Q = 0 \tag{157}$$

produces

$$ax - by = \Gamma, \tag{158a}$$

$$x + (a+q)y - bz = 0, \tag{158b}$$

$$y = -qz. \tag{158c}$$

Using $z = -y/q$ in Eq. (158b) gives

$$y = -\frac{xq}{q(a+q)+b}. \tag{159}$$

Substitution into Eq. (158a) then yields

$$x = \Gamma \frac{q(a+q)+b}{(a+q)(aq+b)}. \tag{160}$$

Because $S(k) = x/\rho_0$, the normalized structure factor is

$$S(k) = \frac{\Gamma}{\rho_0} \frac{q(a+q)+b}{(a+q)(aq+b)}. \tag{161}$$

Substituting the original variables and cancelling the common factor k^2 gives

$$\boxed{S(k) = D\,\frac{\alpha\rho_0 k^2 + (D_c k^2 - \mu)\left[(D+D_c)k^2 - \mu\right]}{\left[(D+D_c)k^2 - \mu\right]\left[D(D_c k^2 - \mu) + \alpha\rho_0\right]}}. \tag{162}$$

The corresponding unnormalized density spectrum is simply

$$C_{\rho\rho}(k) = \rho_0 S(k). \tag{163}$$

Equation (162) immediately gives $S(k) = 1$ for $\alpha = 0$, confirming the Poisson normalization, and it approaches unity at large k. For fixed $\mu < 0$,

$$\lim_{k\to 0} S(k) = \frac{D}{D - \dfrac{\alpha\rho_0}{\mu}}. \tag{164}$$

For $\mu = 0$, the separate expression in Eq. (70) must be used when discussing the long-wavelength limit.

Equation (162) is valid only when both eigenvalues of $A(k)$ have negative real parts. The stability conditions are

$$a + q > 0, \qquad aq + b > 0, \tag{165}$$

which are equivalent to

$$(D + D_c)k^2 - \mu > 0, \qquad D(D_c k^2 - \mu) + \alpha\rho_0 > 0 \tag{166}$$

for nonzero k. If these conditions fail, the Lyapunov equation does not describe a stationary positive-definite covariance, and a steady-state structure factor is not defined within the linear theory.

4.5 Large-argument asymptotics of the integrated sensing kernel

For the sensing kernel in Eq. (87), define

$$F(x) = \int_0^x J_0(u)\,\mathrm{d}u. \tag{167}$$

We show that $F(x) = 1 + O(x^{-1/2})$ as $x \to \infty$, and identify the leading oscillatory correction. The standard improper integral of the Bessel function is

$$\int_0^\infty J_0(u)\,\mathrm{d}u = 1. \tag{168}$$

For example, its value follows from the Laplace transform

$$\int_0^\infty e^{-su} J_0(u)\,\mathrm{d}u = \frac{1}{\sqrt{1+s^2}}, \qquad s > 0, \tag{169}$$

by taking $s \to 0^+$. The undamped improper integral converges by the oscillatory large-u expansion below, so its Abel limit agrees with its ordinary improper value. Consequently,

$$F(x) = 1 - \int_x^\infty J_0(u)\,\mathrm{d}u. \tag{170}$$

The large-argument expansion of J_0 is

$$J_0(u) = \sqrt{\frac{2}{\pi u}}\left[\cos\left(u - \frac{\pi}{4}\right) + \frac{1}{8u}\sin\left(u - \frac{\pi}{4}\right) + O(u^{-2})\right]. \tag{171}$$

Integration by parts gives

$$\begin{aligned}\int_x^\infty u^{-1/2}\cos\left(u - \frac{\pi}{4}\right)\,\mathrm{d}u = &\; -x^{-1/2}\sin\left(x - \frac{\pi}{4}\right) \\ &+ \frac{1}{2}\int_x^\infty u^{-3/2}\sin\left(u - \frac{\pi}{4}\right)\,\mathrm{d}u.\end{aligned} \tag{172}$$

A second integration by parts shows that the last integral is $O(x^{-3/2})$. The sine term in Eq. (171) therefore contributes only $O(x^{-3/2})$ to the tail, while the remaining $O(u^{-5/2})$ term has an absolutely convergent tail of the same order. Hence

$$\int_x^\infty J_0(u)\,\mathrm{d}u = -\sqrt{\frac{2}{\pi x}}\sin\left(x - \frac{\pi}{4}\right) + O(x^{-3/2}). \tag{173}$$

Substitution into Eq. (170) yields

$$\boxed{F(x) = 1 + \sqrt{\frac{2}{\pi x}}\sin\left(x - \frac{\pi}{4}\right) + O(x^{-3/2}) = 1 + O(x^{-1/2}).} \tag{174}$$

Thus $F(x)$ approaches unity through oscillations with a leading amplitude proportional to $x^{-1/2}$, rather than monotonically. Equivalently, for $kd_c \gg 1$,

$$\widehat{K}_{d_c}(k) = \frac{1}{k}\left[1 + \sqrt{\frac{2}{\pi k d_c}}\sin\left(kd_c - \frac{\pi}{4}\right) + O((kd_c)^{-3/2})\right]. \tag{175}$$

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